The 3-adic collision law for the Collatz terminal decoder

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Let

$$2^{a_t} x_{t+1} = 3x_t + 1, \qquad a_t = v_2(3x_t + 1), \tag{1}$$

where every x_t is a positive odd integer. Put

$$Y_t := x_t + 1, \qquad b_t := v_3(Y_t), \qquad Y_t = 3^{b_t} u_t, \qquad 3 \nmid u_t. \tag{2}$$

For $a \geq 1$, define

$$r(a) := 2^a - 2, \qquad q(a) := v_3(r(a)), \tag{3}$$

where $q(1) := \infty$.

Lemma 1 (valuation of the decoder forcing term)**.** *For every $a \geq 1$,*

$$q(a) = \begin{cases} \infty, & a = 1, \\ 0, & a \text{ is even}, \\ 1 + v_3\big((a-1)/2\big), & a > 1 \text{ is odd}. \end{cases} \tag{4}$$

Proof. The first two cases follow directly from $r(1) = 0$ and $2^a \equiv 1 \pmod 3$ when a is even. Suppose that $a = 2n + 1 > 1$. Then

$$r(a) = 2(4^n - 1). \tag{5}$$

Write $n = 3^s m$, where $3 \nmid m$. The binomial theorem gives

$$4^m - 1 = (1+3)^m - 1 \equiv 3m \pmod 9,$$

so $v_3(4^m - 1) = 1$. If $X \equiv 1 \pmod 3$, then $X^3 - 1 = (X-1)(X^2 + X + 1)$, while $X^2 + X + 1 \equiv 3 \pmod 9$. Thus replacing an exponent by three times that exponent raises the 3-adic valuation by exactly one. Repeating this s times gives

$$v_3(4^n - 1) = 1 + s = 1 + v_3(n),$$

which proves (4). □

Theorem 2 (decoder collision law)**.** *Fix one step of (1), and abbreviate*

$$a := a_t, \qquad b := b_t, \qquad b' := b_{t+1}, \qquad u := u_t.$$

Then

$$\boxed{2^a 3^{b'} u_{t+1} = 3^{b+1} u + r(a).} \tag{6}$$

If $q(a) \neq b+1$*, then*

$$\boxed{b' = \min\{q(a), b+1\}.} \tag{7}$$

If $q(a) = b+1 < \infty$*, put*

$$\rho_a := \frac{r(a)}{3^{q(a)}}. \tag{8}$$

Then $3 \nmid \rho_a$ *and*

$$\boxed{b' = b + 1 + v_3(u + \rho_a).} \tag{9}$$

In particular,

$$a = 1 \Longrightarrow b' = b+1, \qquad a \equiv 0 \pmod 2 \Longrightarrow b' = 0. \tag{10}$$

Proof. Adding $2^a - 1$ to both sides of (1) gives

$$2^a(x_{t+1} + 1) = 3(x_t + 1) + 2^a - 2.$$

Substitution from (2) proves (6). Since 2^a is a unit modulo 3, the number b' is the valuation of the right side of (6).

The two summands on that side have valuations $b+1$ and $q(a)$. If these valuations are distinct, the valuation of their sum is their minimum. This proves (7). If they are equal and finite, factoring out 3^{b+1} gives

$$3^{b+1}(u + \rho_a),$$

which proves (9). The two statements in (10) follow from (4), (7), and $q(1) = \infty$. □

Corollary 3 (bounded-ceiling exponent sieve)**.** *Fix* $D \geq 1$*, and suppose that* $0 \leq b \leq D$*. The condition* $b' \leq D$ *is characterized as follows.*

1. *If* $b = D$*, then*

$$\boxed{b' \leq D \quad \Longleftrightarrow \quad a \not\equiv 1 \pmod{2 \cdot 3^D}.} \tag{11}$$

2. If $b < D$ and $q(a) = b + 1$, then

$$\boxed{b' \le D \quad \Longleftrightarrow \quad u \not\equiv -\rho_a \pmod{3^{D-b}}.} \tag{12}$$

3. If $b < D$ and $q(a) \neq b + 1$, then $b' \le D$ automatically.

Consequently, if $b_t \le D$ on a tail, then every time with $b_t = D$ satisfies

$$a_t \not\equiv 1 \pmod{2 \cdot 3^D}, \tag{13}$$

and that tail contains no factor 1^{D+1}.

Proof. Suppose first that $b = D$. If $q(a) \ge D + 1$, equations (7) and (9) give $b' \ge D + 1$. If $q(a) \le D$, equation (7) gives $b' = q(a) \le D$. Therefore,

$$b' \le D \quad \Longleftrightarrow \quad q(a) \le D. \tag{14}$$

By (4), the complementary condition $q(a) \ge D + 1$ is equivalent to

$$a \equiv 1 \pmod{2 \cdot 3^D},$$

including the case $a = 1$. This proves (11).

Now suppose that $b < D$. If $q(a) = b + 1$, equation (9) gives

$$b' \le D \quad \Longleftrightarrow \quad v_3(u + \rho_a) \le D - b - 1.$$

This is exactly (12). If $q(a) < b + 1$, then (7) gives $b' = q(a) \le b < D$. If $q(a) > b + 1$, the same equation gives $b' = b + 1 \le D$. This proves the third assertion.

Finally, equation (13) is (11) applied along the tail. Each exponent-one step raises the decoder depth by one according to (10). A factor 1^{D+1} would therefore raise a nonnegative decoder depth above D, which is impossible. □

Theorem 4 (one-forbidden-class decoder automaton)**.** *Fix $D \ge 0$, and put*

$$M_D := 3^{D+1}, \qquad H_D := 2 \cdot 3^D. \tag{15}$$

For $s \in \mathbb{Z}/M_D\mathbb{Z}$ and $a \ge 1$, define

$$F_a(s) := 2^{-a}\big(3s + 2^a - 2\big) \pmod{M_D}. \tag{16}$$

If $s_t \equiv x_t + 1 \pmod{M_D}$, then

$$s_{t+1} = F_{a_t}(s_t). \tag{17}$$

Moreover, for every nonzero state s, there is exactly one residue class $\chi_D(s)$ (mod H_D) such that

$$\boxed{F_a(s) = 0 \quad \Longleftrightarrow \quad a \equiv \chi_D(s) \pmod{H_D}.} \tag{18}$$

Equivalently, at every state with decoder depth at most D, exactly one exponent class modulo $2 \cdot 3^D$ raises the next decoder depth above D.

Proof. Equation (17) is (6) reduced modulo M_D. We next verify that 2 generates the unit group modulo 3^k for every $k \geq 1$. The calculation in the proof of Lemma 1 gives

$$v_3(2^{2m} - 1) = v_3(4^m - 1) = 1 + v_3(m). \tag{19}$$

An exponent h satisfying $2^h \equiv 1 \pmod{3^k}$ must be even, say $h = 2m$, and (19) shows that its least positive value is $2 \cdot 3^{k-1}$. This equals $\varphi(3^k)$, so 2 is a generator.

Now $F_a(s) = 0$ is equivalent to

$$2^a \equiv 2 - 3s \pmod{M_D}. \tag{20}$$

The right side is a unit, and the powers of 2 run once through all units as a runs through a complete residue system modulo H_D. Hence (20) selects exactly one exponent class, proving (18). □

Theorem 5 (exact finite realization of decoder paths)**.** *Fix $D \geq 0$ and a nonzero initial state $s_0 \in \mathbb{Z}/M_D\mathbb{Z}$. Let $(a_0, \ldots, a_{L-1})$ be any positive exponent word, and define recursively*

$$s_{t+1} := F_{a_t}(s_t). \tag{21}$$

If $s_t \neq 0$ for $0 \leq t \leq L$, then infinitely many positive odd integers n have this exact accelerated exponent prefix and satisfy

$$x_t + 1 \equiv s_t \pmod{M_D} \qquad (0 \leq t \leq L). \tag{22}$$

In particular, their decoder depths obey $v_3(x_t + 1) \leq D$ throughout the displayed finite orbit segment.

If $D \geq 1$ and the exponent alphabet is

$$\mathcal{A}_D := \{1, 2, \ldots, H_D\}, \tag{23}$$

then exactly $(H_D - 1)^L$ words in $\mathcal{A}_D^L$ generate a nonzero path from any prescribed nonzero state s_0. Every one of those words has the positive realizations asserted above.

Proof. We first recall the finite cylinder fact and include its induction. A positive exponent word of total exponent A is realized exactly by one odd residue class modulo 2^{A+1}. The empty word gives the unique odd class modulo 2. Suppose that a prefix of length t and total exponent A has representative ρ. Replacing its initial value by $\rho+\ell 2^{A+1}$ replaces its endpoint by

$$x_t(\rho) + 2\ell 3^t. \tag{24}$$

To append the exponent a, exactness is equivalent to

$$3x_t(\rho) + 1 + 6\ell 3^t \equiv 2^a \pmod{2^{a+1}}. \tag{25}$$

After division by 2, the coefficient of ℓ is the odd number 3^{t+1}. Thus (25) selects one class $\ell \pmod{2^a}$, completing the induction.

Let ρ be the resulting cylinder class for the displayed word and put $A := \sum_{t=0}^{L-1} a_t$. The simultaneous congruences

$$n \equiv \rho \pmod{2^{A+1}}, \qquad n \equiv s_0 - 1 \pmod{M_D} \tag{26}$$

have one class modulo $2^{A+1}M_D$ by the Chinese remainder theorem. This class contains infinitely many positive odd integers. Its first congruence gives the exact exponent word. Its second congruence, followed by (17), proves (22) by induction.

Finally, the alphabet (23) contains one representative of every exponent class modulo H_D. Theorem 4 says that exactly one of these H_D letters leaves the nonzero state space at each step. Hence the rooted path tree has exactly $H_D - 1$ children at every vertex and exactly $(H_D - 1)^L$ paths of length L. The first part of the theorem realizes each path. □